Intrinsic Volume-Error Certificates for Hypersymplectic Deformation

A source-specific study of auxiliary forms and collapsing nodes

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Abstract

We study the auxiliary prescribed-volume step in the manuscript *Deforming hypersymplectic four-manifolds to hyperkähler triples*. For two closed, cohomologous, positive forms orthogonal to the same definite pair, we prove an exact midpoint-metric energy identity and a sharp square-root density estimate. The identity retains the cohomological defect when the classes differ and quantifies the cost of returning to a prescribed metric. We also optimize the quadratic majorant in the source's exact-form space, retaining its three moment conditions, positivity margin, and fixed-order collapsing estimates. An explicit annular Fourier remainder separates nodal gluing accuracy from the global completion of a singular fibre. Closed torus examples show sharpness and the failure of omitted cohomology or metric conditions. Classical negative-chord uniqueness, Hodge theory, and contraction arguments are credited; the source's current, family, lattice, pencil, and endpoint existence machinery is not independently certified by these local estimates. Contribution assessment: C3/5, sharp intrinsic volume-error bounds and an enlarged exact-form correction certificate provide a notable local refinement while leaving the global deformation obligations explicit.

1 Problem, normalization, and claim boundary

Let X be a closed connected oriented smooth four-manifold. Products of two-forms mean wedge products. A triple of real closed two-forms is positive if its span is a positive three-plane for the wedge pairing at every point. The source normalizes the cohomological Gram matrix by $\int_X \omega_i\omega_j = \delta_{ij}$. Its claimed endpoint satisfies $\omega_i\omega_j = 2\delta_{ij}\nu$, so $\int_X \nu = 1/2$. This is not the convention $\int_X \nu = 1$. A constant real change of coefficient basis handles a general positive cohomological Gram matrix; it is not a variable gauge transformation.

The source [1] claims a smooth positive deformation with each of the three individual classes fixed. It constructs an auxiliary closed form D for a prepared definite pair A, C:

$$AD = CD = 0, \qquad D^2 = A^2, \qquad (A, C, D) \text{ positive.} \tag{1.1}$$

The auxiliary class is generally *not* the original third class $[B]$. A separate uniform curve-chamber argument keeps $[B]$ available, and two Kähler prescribed-volume steps construct the actual endpoint. Solving (1.1) locally therefore does not by itself prove the global deformation theorem.

We retain the source's geometric domain. A positive three-plane in $\Lambda^2 T^*X$, together with a positive volume form ν, determines a Riemannian metric whose self-dual plane is that three-plane and whose volume is ν. Write

$$Q(\xi,\zeta) = \frac{\xi\zeta}{2\nu}.$$

This fibrewise bilinear form has signature $(3,3)$. The orthogonal complement of a definite pair has signature $(1,3)$; one must choose one of its two timelike cones. In the estimates below $\nu = A^2/2$, but $C^2 = A^2$ and $AC = 0$ are not assumed unless expressly stated.

What is established here. Theorems 2.1, 3.1, and 4.1 are local quantitative statements with complete proofs. They require actual closed forms or an actual continuous right inverse on specified spaces, not a finite matrix standing in for those objects. Proposition 5.1 gives a finite-order annular error certificate. The full source architecture is mapped in Section 7; reading its text is distinct from independently proving its deep global inputs.

2 Midpoint energy and square-root densities

Fix smooth closed A, C forming a definite pair. Suppose D_0, D_1 are smooth closed forms satisfying $AD_i = CD_i = 0$, with positive square and in the same timelike cone of $\operatorname{span}(A,C)^\perp$. Set

$$h = D_1 - D_0, \quad H = \frac{D_0 + D_1}{2}, \quad q_i = \frac{D_i^2}{2\nu}, \quad q_H = \frac{H^2}{2\nu}. \tag{2.1}$$

Then $q_i, q_H > 0$. Let g_H have volume ν and self-dual plane $\operatorname{span}(A,C,H)$. The density norm

$$\mathcal{H}_\nu(q_0,q_1)^2 = \int_X (\sqrt{q_1} - \sqrt{q_0})^2 \, \nu$$

is an unnormalized square-root distance. No probability-mass normalization is implicit.

Theorem 2.1 (Exact energy and a sharp density bound)**.** *Under these hypotheses,*

$$\|h\|^2_{L^2(g_H)} = \int_X \frac{(q_1 - q_0)^2}{q_H}\,\nu - \int_X h^2. \tag{2.2}$$

In particular, if $[D_1] = [D_0]$*, then*

$$\|D_1 - D_0\|^2_{L^2(g_H)} = \int_X \frac{(q_1 - q_0)^2}{q_H}\,\nu \le 4\mathcal{H}_\nu(q_0,q_1)^2. \tag{2.3}$$

The coefficient 4 *cannot be decreased for all smooth closed cohomologous positive forms in this setting. The result imposes no lower bound on a one-form spectral gap.*

Proof. Because $Ah = Ch = 0$, the self-dual component of h is a multiple of H. Moreover

$$Hh = \frac{D_1^2 - D_0^2}{2} = (q_1 - q_0)\nu, \qquad |H|^2_{g_H} = 2q_H.$$

Consequently

$$h^+ = \frac{q_1 - q_0}{2q_H} H, \qquad |h^+|^2_{g_H} = \frac{(q_1 - q_0)^2}{2q_H}.$$

For any two-form, $|h|^2\nu = 2|h^+|^2\nu - h^2$. Integration proves (2.2). If $h = \mathrm{d}a$, Stokes' theorem on the closed manifold gives $\int_X h^2 = \int_X \mathrm{d}(a\mathrm{d}a) = 0$.

The reverse Cauchy–Schwarz inequality in a Lorentz space, for vectors in the same timelike cone, gives $Q(D_0, D_1) \geq \sqrt{q_0 q_1}$. To see this directly, take $D_0 = (\sqrt{q_0}, 0)$ in a Lorentz orthonormal basis; then $D_1 = (t, z)$, $t > 0$, $t^2 - |z|^2 = q_1$, and $t \geq \sqrt{q_1}$. Thus

$$q_H = \frac{q_0 + q_1 + 2Q(D_0, D_1)}{4} \geq \frac{(\sqrt{q_0} + \sqrt{q_1})^2}{4}.$$

Cancelling the square of $\sqrt{q_0} + \sqrt{q_1}$ proves (2.3). Sharpness is proved by the closed-form family in Example 3.2, not merely by independent pointwise vectors. □

Corollary 2.2 (Conditional a posteriori control)**.** *Suppose an actual solution D of (1.1) exists in the same class and cone as a closed approximation $\widehat{D}$ with $A\widehat{D} = C\widehat{D} = 0$. Put $\widehat{q} = \widehat{D}^2/(2\nu) > 0$. In their midpoint metric,*

$$\left\|\widehat{D} - D\right\|_2 \leq 2\left\|\sqrt{\widehat{q}} - 1\right\|_{L^2(\nu)}.$$

If $\widehat{q} \geq \mu > 0$, this is at most

$$\frac{2}{1 + \sqrt{\mu}} \left\|\widehat{q} - 1\right\|_{L^2(\nu)}. \tag{$*$}$$

These estimates compare existing forms; they do not establish existence of D, exactness of a numerical field, or a C^2 bound.

Proof. Use $q_0 = 1$ in (2.3), then $|\sqrt{\widehat{q}} - 1| = |\widehat{q} - 1|/(\sqrt{\widehat{q}} + 1)$. □

Uniqueness is not the new claim. If $q_0 = q_1$ and the classes agree, (2.3) gives $D_0 = D_1$. This is the prescribed-volume negative-chord mechanism already treated by Donaldson [2, Section 2, Proposition 1, and Section 3]. The contribution here is the quantitative midpoint identity, its sharp density bound and explicit norm-transfer cost in this auxiliary equation. No historical priority is asserted. On connected X, equal classes also exclude globally opposite cones: opposite timelike vectors have $D_0 D_1 < 0$ everywhere, whereas $\int_X D_0 D_1 = \int_X D_0^2 > 0$.

3 Physical metric conversion and closed examples

The midpoint metric is intrinsic, but depends on both forms. Its identification with a fixed computational metric has a real cost. Let g_0 have volume ν and self-dual plane $\mathrm{span}(A, C, D_0)$. Define the nonnegative angle $\xi(x)$ by

$$\cosh \xi = \frac{Q(D_0, H)}{\sqrt{q_0 q_H}}. \tag{3.1}$$

Theorem 3.1 (Cone-angle cost)**.** *For every real two-form ζ, pointwise,*

$$e^{-2\xi}|\zeta|^2_{g_H} \leq |\zeta|^2_{g_0} \leq e^{2\xi}|\zeta|^2_{g_H}.$$

Hence if $\xi \leq \Xi$ on X and $[D_0] = [D_1]$,

$$\|D_1 - D_0\|_{L^2(g_0)} \leq 2e^{\Xi}\mathcal{H}_\nu(q_0, q_1). \tag{3.2}$$

A uniform comparison requires a uniform angle bound; scalar volume-density control alone is not such a bound.

Proof. Choose a Q-orthonormal basis for the shared positive pair-plane. In its Lorentz complement write $e_0 = D_0/\sqrt{q_0}$ and $H/\sqrt{q_H} = \cosh\xi\, e_0 + \sinh\xi\, e_1$, with $Q(e_1, e_1) = -1$; when $\xi = 0$ choose any negative unit vector. The corresponding negative vector is $\sinh\xi\, e_0 + \cosh\xi\, e_1$. On this two-plane the two positive Hodge norms differ by the Lorentz boost matrix, whose Euclidean singular values are e^{ξ} and $e^{-\xi}$. On the other four directions the norms coincide. Both Hodge norms have the same factor 2 relative to these Q-coordinates, because the volume is the same. This proves the pointwise comparison. Integrate and apply Theorem 2.1. □

Example 3.2 (Closed torus family and sharp coefficient)**.** Use coordinates of period 1 on T^4, write $e^{ij} = \mathrm{d}x_i \wedge \mathrm{d}x_j$, and set

$$\nu = e^{1234}, \quad A = e^{13} - e^{24}, \quad C = e^{14} + e^{23}, \quad D_0 = e^{12} + e^{34}, \quad D_1 = (1 + f(x_1))e^{12} + e^{34}.$$

Let f be smooth, periodic, mean zero and strictly greater than -1. These are actual closed forms, with $h = \mathrm{d}(F(x_1)\mathrm{d}x_2)$ for a periodic primitive $F' = f$. They are orthogonal to A, C and in the same positive cone. Here

$$q_0 = 1, \quad q_1 = 1 + f, \quad q_H = 1 + f/2, \quad g_H = \sqrt{q_H}(\mathrm{d}x_1^2 + \mathrm{d}x_2^2) + q_H^{-1/2}(\mathrm{d}x_3^2 + \mathrm{d}x_4^2).$$

In particular

$$\|h\|^2_{L^2(g_H)} = \int_{T^4} \frac{f^2}{1 + f/2}\nu. \tag{3.3}$$

Take $f = a\cos(2\pi x_1)$, $0 < |a| < 1$. Dividing both sides of (2.3) by a^2, dominated convergence gives respective limits $1/2$ and $4(1/8) = 1/2$. Thus $\|h\|_2^2 / \mathcal{H}^2_\nu \to 4$. This example has volume 1, harmless for the local theorem. Multiplying all two-forms by $1/\sqrt{2}$ gives the source's cohomological normalization and volume $1/2$.

Example 3.3 (Cohomology cannot be discarded)**.** On the same torus take $D_\lambda = \lambda e^{12} + \lambda^{-1}e^{34}$, $\lambda > 0$. It has the same volume density 1 as D_0, but generally a different class. For $h = D_\lambda - D_0$,

$$\int h^2 = 2(2 - \lambda - \lambda^{-1}), \qquad \|h\|^2_{L^2(g_H)} = 2(\lambda + \lambda^{-1} - 2).$$

The missing topological term accounts for the entire nonzero error. As $\lambda \to \infty$, equal densities also coexist with unbounded cone angle. Neither a scalar residual nor a finite list of unchecked periods eliminates this example.

A collapsing-coordinate warning. For $D_\rho = \rho e^{12} + \rho^{-1}e^{34}$, an exact perturbation $\eta f(x_1)e^{12}$ changes the density by $\eta f/\rho$, not ηf. Its midpoint norm is $\int (\eta/\rho)^2 f^2/(1 + \eta f/(2\rho))\,\nu$. Thus small coordinate coefficients in a fixed atlas need not be small in the physical collapsing metric. The source's intrinsic Sobolev norms cannot be silently replaced by unweighted coefficient norms.

4 Exact-form correction with the optimal quadratic radius

We now return to the actual approximate triple (A, C, D^0_ε) in source Section 10. Fix $\varepsilon > 0$ and an integer $k \geq 6$. All Sobolev norms in this section are intrinsic to its metric g_ε, with volume $\nu = A^2/2$. Let

$$\mathcal{E}^k = \{h \in H^k(\Lambda^2) : \mathrm{d}h = 0, \text{ harmonic projection of } h = 0\}.$$

This is a closed Banach subspace; for the fixed smooth metric its members are exact and have H^{k+1} primitives. Let $\mathcal{Z}^k$ be triples of H^k top-forms, each with integral zero.

4.1 The continuous inverse and its three moments

The source establishes $b^+(X) = 3$ by its global topology argument. Retain this as an assumption here. Put $E = (A, C, D^0_\varepsilon)$, and let $M_{ij} = \langle E_i, E_j \rangle_{g_\varepsilon}$, the pointwise physical Gram matrix. For $f_i = t_i \nu$, the desired self-dual part is

$$\phi = \sum_j (M^{-1}t)_j E_j, \qquad \langle \phi, E_i \rangle_{L^2} = \int_X f_i = 0. \tag{4.1}$$

The three E_i are a basis of the harmonic self-dual forms. Hodge theory on the fixed closed metric therefore supplies a Green operator G and

$$h = 2\mathrm{d}\mathrm{d}^* G\phi, \qquad h^+ = \phi, \qquad \|h\|_2^2 = 2\,\|\phi\|_2^2 . \tag{4.2}$$

The last equality is Stokes, not an estimate of $\|G\|$. An exact anti-self-dual form has zero norm, proving uniqueness. Thus the continuous inverse

$$\mathcal{L} : \mathcal{Z}^k \longrightarrow \mathcal{E}^k, \qquad (Ah, Ch, D^0_\varepsilon h) = f$$

is well defined. Higher derivative bounds use $\mathrm{d}h = 0$, $\mathrm{d}^* h = -2 * \mathrm{d}\phi$, Weitzenböck identities and curvature derivatives; the source proves a bound $P_\varepsilon \le C_k \varepsilon^{-p_k}$. This polynomial estimate is an inherited geometric input, not a consequence of inverting $M(x)$ at finitely many points. The inverse is for the two-form h, not a unique primitive a.

The source first removes the two linear periods of its glued form, then rescales its square period. In its normalization,

$$a = \int EA, \quad b = \int EC, \quad \widehat{E} = E - aA - bC, \quad q = \int \widehat{E}^2, \quad D^0_\varepsilon = q^{-1/2}\widehat{E}.$$

Thus $\int AD^0_\varepsilon = \int CD^0_\varepsilon = 0$ and $\int (D^0_\varepsilon)^2 = \int A^2 = 1$. Exact corrections cannot repair these three scalar discrepancies after the fact.

4.2 Quadratic majorant and positivity

Suppose, on these actual spaces,

$$\|\mathcal{L}f\|_{H^k} \le P\,\|f\|_{H^k}, \qquad \left\|(0, 0, h\widetilde{h})\right\|_{H^k} \le Q\,\|h\|_{H^k}\left\|\widetilde{h}\right\|_{H^k}, \tag{4.3}$$

where $P, Q > 0$; the product norm is the same one used in the three-slot inverse. Write

$$\mathcal{R} = (-AD^0_\varepsilon, -CD^0_\varepsilon, (A^2 - (D^0_\varepsilon)^2)/2), \qquad \delta = \|\mathcal{R}\|_{H^k} .$$

Each component has zero integral. For every exact h, $\int h^2 = 0$, so

$$T(h) = \mathcal{L}\{\mathcal{R} - (0, 0, h^2/2)\} \tag{4.4}$$

never leaves the inverse's compatible data space.

Theorem 4.1 (Enlarged exact-form correction certificate)**.** *Assume* (4.3) *and* $\sigma = 2P^2Q\delta < 1$. *Define*

$$r_* = \frac{2P\delta}{1 + \sqrt{1 - \sigma}}, \qquad q_* = PQr_* = 1 - \sqrt{1 - \sigma}. \tag{4.5}$$

For $\delta = 0$ take $r_* = q_* = 0$. Then T has a unique fixed point in $\overline{B}_{\mathcal{E}^k}(0, r_*)$. It satisfies all three equations (1.1), except that positivity must separately be certified as follows: if

$$\|h\|_{C^0(g_\varepsilon)} \le S_\varepsilon \|h\|_{H^k} \quad \text{and} \quad S_\varepsilon r_* < m_\varepsilon,$$

where $m_\varepsilon > 0$ is an actual C^0 perturbation radius preserving the positive component of (A, C, D^0_ε), the solution is positive. Under the source's smooth-background elliptic bootstrap it is smooth.

Proof. The smaller nonnegative root of

$$r = P\delta + \tfrac{1}{2}PQr^2$$

is (4.5). On its ball, (4.3) gives $\|T(h)\| \le P\delta + PQr_*^2/2 = r_*$. Since two-forms commute under wedge product,

$$\left\|T(h) - T(\widetilde{h})\right\| \le \tfrac{1}{2}PQ(\|h\| + \left\|\widetilde{h}\right\|)\left\|h - \widetilde{h}\right\| \le q_* \left\|h - \widetilde{h}\right\|.$$

Here $q_* < 1$. The ball is complete, so contraction proves existence and uniqueness there. Applying the defining continuous inverse to (4.4) gives $A(D^0_\varepsilon + h) = C(D^0_\varepsilon + h) = 0$ and $(D^0_\varepsilon + h)^2 = A^2$. Positivity follows from the separate embedding and open-cone margin, not from the quadratic discriminant.

For smoothness, choose a Coulomb primitive $h = \mathrm{d}a$, $\mathrm{d}^* a = 0$ for this fixed metric. At a nonzero covector ζ, a vector b in the symbol kernel has $\zeta \wedge b$ wedge-orthogonal to a positive three-plane and has $(\zeta \wedge b)^2 = 0$. The negative-definite complement forces $\zeta \wedge b = 0$; the gauge contraction then forces $b = 0$. This is the source's square elliptic symbol. Fixed-metric local regularity bootstraps the initial H^{k+1} primitive to smoothness. It does not assert uniform estimates over all derivative orders as $\varepsilon \downarrow 0$. □

Exact increment and cost. The source explicitly imposes $\sigma < 1/2$ with ball $2P\delta$. The theorem permits $\sigma < 1$, with a smaller ball $r_* \le 2P\delta$; it optimizes that scalar majorant, not the underlying inverse. As $\sigma \uparrow 1$, the contraction factor tends to 1, so the larger admissible range sacrifices iteration speed. At $\sigma = 1$ this argument proves no strict contraction. For Picard iterates starting at zero,

$$\|h_* - h_n\| \le q_*^n r_*, \qquad \|h_* - h_n\| \le \frac{\|h_{n+1} - h_n\|}{1 - q_*}.$$

The latter assumes exact evaluation of T, or separately bounded evaluation error. Scalar discriminant optimization is classical; its value here is the explicit finite-ε improvement while preserving exactness, the mean conditions and physical norms.

Example 4.2 (A rational threshold fixture)**.** Take certified bounds $P = 2, Q = 3, \delta = 5/128$. Then $\sigma = 15/16$, $r_* = 1/8$, and $q_* = 3/4$. Indeed $5/64 + 3/64 = 1/8$. The source's displayed $\sigma < 1/2$ test fails, while this quadratic certificate passes. One must still verify $S_\varepsilon/8 < m_\varepsilon$ and that these are genuine infinite-dimensional bounds; these numbers are a worked certificate, not measured constants for an actual collapsing K3 surface.

Order of choices. The source proves, at each fixed finite order,

$$P_\varepsilon \le C\varepsilon^{-p}, \quad Q_\varepsilon \le C\varepsilon^{-q}, \quad \delta_\varepsilon \le C\varepsilon^{-s}e^{-c/\varepsilon}.$$

Hence $\sigma_\varepsilon \to 0$; the embedding cost is polynomial as well. The source already uses this domination. Our relaxed test changes which certified finite parameters pass, not the existence of that asymptotic domination, the topology, or the final chamber argument.

5 A bounded annular Fourier certificate

The source's nodal model is not merely a formal Fourier solution. It completes a circle bundle of linking Chern number -1, with circle period 1, at a unit monopole; the model has a section and is identified with the prescribed holomorphic elliptic family. For the unit holomorphic normalization and fibre area $\rho > 0$, its potential on $0 < |s| < 1$, $t \in \mathbb{R}/\rho\mathbb{Z}$, is

$$V_\rho(s,t) = \frac{\tau_2(s)}{\rho} + \frac{1}{2\pi\rho}\sum_{n\neq 0} e^{2\pi i n t/\rho} K_0(2\pi|n||s|/\rho), \qquad \tau_2 = -\frac{\log|s|}{2\pi}. \tag{5.1}$$

Fix $0 < r_- < r_+ < 1$ before choosing ρ. Let $V_{\rho,M}$ retain $|n| \le M$, $M \ge 0$. The mean in t is exactly τ_2/ρ at every truncation.

Proposition 5.1 (Explicit zeroth-order tail)**.** *On $r_- \le |s| \le r_+$, put $a = 2\pi r_-/\rho$. Then*

$$\|V_\rho - V_{\rho,M}\|_\infty \le B_M := \frac{e^{-a(M+1)}}{2\pi\sqrt{\rho r_-}\sqrt{M+1}(1-e^{-a})}. \tag{5.2}$$

In particular $\tau_{\min}/\rho - B_0 > 0$, with $\tau_{\min} = -\log r_+/(2\pi)$, suffices for positivity of the full potential and every truncated potential on that annulus.

Proof. The classical integral formula [6, 10.32.9], $K_0(x) = \int_0^\infty e^{-x\cosh u}\mathrm{d}u$, and $\cosh u \ge 1 + u^2/2$ imply

$$0 < K_0(x) \le e^{-x}\sqrt{\frac{\pi}{2x}} \quad (x > 0).$$

Pair the positive and negative frequencies in (5.1), take absolute values and obtain

$$\frac{1}{2\pi\sqrt{\rho r_-}}\sum_{n=M+1}^{\infty} n^{-1/2}e^{-an}.$$

Bound $n^{-1/2} \le (M+1)^{-1/2}$ and sum the geometric series. For positivity, the absolute sum of any subset of nonzero frequencies is bounded by B_0. □

Example 5.2 (An annulus fixed before collapse)**.** Take $r_- = 1/4, r_+ = 1/2, \rho = 1/32$. Then $a = 16\pi$ and

$$\frac{\rho B_0}{\tau_{\min}} = \frac{1}{2\sqrt{2}\log 2}\frac{e^{-16\pi}}{1-e^{-16\pi}} < 2^{-47}.$$

Use $\pi > 3,\ e > 2,\ \log 2 > 1/2$ to obtain the last elementary bound. This certifies positivity and a small relative tail on the named annulus. It says nothing about the unresolved global gluing or the singular point $s = 0$.

Derivative and core limitations. Differentiating j times in t inserts $(2\pi n/\rho)^j$. Thus the same proof, for every fixed j, gives the explicit convergent series bound

$$\left\|\partial_t^j(V_\rho - V_{\rho,M})\right\|_\infty \le \frac{(2\pi/\rho)^j}{2\pi\sqrt{\rho r_-}}\sum_{n>M} n^{j-1/2}e^{-an}.$$

Radial derivatives and connection primitives need their own bounds. A zeroth-order bound cannot be inserted into the source's H^k contraction test without them. For general holomorphic coefficient

$h \neq 0$, the source uses $\rho = \varepsilon/|h|$ and rescales the triple and metric by $|h|$; forgetting this changes the fibre area.

Finite M is not a replacement for the completed nodal metric. Indeed $K_0(x) = O(1 + |\log x|)$ as $x \downarrow 0$, whereas the source's full potential has the three-dimensional monopole singularity $1/(4\pi\sqrt{|s|^2 + t^2})$. The logarithmic bound follows from its integral formula by splitting at $u = \log(2/x)$ and using $\cosh u \geq e^u/2$ on the tail. A finite sum has only logarithmic growth at $t = 0$, not $1/|s|$. Hence a finite annular computation does not certify the Hopf completion, ordinary-node topology, or the required two annular periods.

6 Ram-compatible observations without changing the PDE

The freshly read original private proof files [7, 8, 9] are used for domain-aware reconstruction and error accounting, not as independent proofs of hypersymplectic topology. Their relevant content is specific. TR8 treats an inverse as an operator with its boundary domain, separates Neumann compatibility from a chosen mean-zero representative, and proves the coercive inverse in its physical weak norm. The stability manuscript proves a contraction certificate for an actual bounded isomorphism and warns that a rectangular left inverse can annihilate a nonzero residual. The jet manuscript proves the Taylor-moment remainder and full Leibniz rows, retaining both moving-endpoint and discarded higher-derivative terms.

6.1 Exact moments and a physical retained space

For smooth exact trial forms $h_j = \mathrm{d}a_j$ on closed X, choose

$$h(c) = \sum_j c_j h_j, \qquad G_{ij} = \int_X \langle h_i, h_j \rangle_{g_\varepsilon} \nu.$$

Then $\|h(c)\|_2^2 = c^{\mathsf{T}} G c$ exactly. Remove any linear dependence before declaring this a norm on coefficient space. All mixed square moments vanish:

$$\int_X h_i h_j = \int_X \mathrm{d}(a_i \mathrm{d}a_j) = 0.$$

Consequently the source's three mean constraints are preserved for every coefficient vector when the initial approximate form has already been moment-corrected. This is more appropriate than fitting an unconstrained pointwise polynomial field and hoping its cohomology is correct.

For any exact h and any positive closed frame E_i spanning the self-dual plane, let $t_i = E_i h/\nu$. The same projection calculation as (4.1) gives the continuous identity

$$\|h\|_2^2 = 2 \int_X t^{\mathsf{T}} M^{-1} t \, \nu. \tag{6.1}$$

For a closed but nonexact h, subtract $\int_X h^2$ on the right. Equation (6.1) is a physical energy representation, not a claim that finitely many observations determine every exact form.

Indeed finitely many linear moments have a nontrivial kernel on the infinite-dimensional space of smooth exact two-forms. Choose more linearly independent compactly supported exact forms in a coordinate ball than there are tests, and solve the resulting homogeneous finite linear system. This gives a nonzero smooth exact field with all those measured moments zero. Finite rank therefore cannot replace a bound for $\mathcal{L} : \mathcal{Z}^k \to \mathcal{E}^k$.

6.2 A local chart error ledger

The jet proof's actual statement on a fixed displacement window Γ is

$$\int_\Gamma K(x,\alpha)f(x-\alpha)\mathrm{d}\alpha = \sum_{n=0}^{N}\frac{(-1)^n}{n!}\mu_n(x)f^{(n)}(x)+\rho_N(x), \qquad \mu_n=\int_\Gamma \alpha^n K(x,\alpha)\mathrm{d}\alpha,$$

with

$$|\rho_N| \le \frac{\left\|f^{(N+1)}\right\|_\infty}{(N+1)!}\int_\Gamma |\alpha|^{N+1}|K(x,\alpha)|\mathrm{d}\alpha.$$

Differentiating k times requires derivatives of K, justified domination, and up to $N+1+k$ derivatives of f for the differentiated-remainder bound. Cutting off the resulting jet at degree N additionally drops the terms $\binom{k}{p}S_n^{(k-p)}f^{(n+p)}$ with $n+p>N$. Neither loss disappears because the forward moment identity is exact on polynomials. Moving windows also produce endpoint terms.

In the nodal application these formulas may be used on a fixed annulus to approximate bounded coefficient integrals. Coordinate Jacobians, derivatives of the physical metric, partition functions, and connection coefficients belong in the actual remainder. One must then assemble a bound in the same intrinsic H^k norm as δ, rather than substitute a maximum sampled scalar error. The source's fixed-order polynomial geometric losses are precisely what make such a transfer plausible; they are not supplied by the moment transform itself. No computational speedup is measured here.

7 Full source architecture and imported obligations

All eleven scientific modules were read literally. The following map records their logical role, not a declaration that every external theorem or global argument has been independently certified.

Module	Source mechanism	Boundary retained here
01–02	Positive triples, definite-pair almost-complex structures, topological restrictions and the endpoint statement.	Normalization is $\int\omega_i\omega_j=\delta_{ij}$. The $b^+=3$ conclusion is an imported global input to the inverse.
03	Closed lift, positive separating currents, resolvent smoothing, radial and angular energy estimates.	Closedness of a current is distinguished from coefficient closedness; the Hodge-star convention and harmonic part remain.
04	Density limits, closed density truncations, rectifiable integral cycles and a taming-cone criterion.	Compatibility precedes the external curve-regularity input. Finite sampling does not recognize an integral curve.
05	Coefficient-sphere family invariants, wall crossing and compact-family Taubes limits through a marked point.	The parameter eigenline twist, based gauge, curvature normalization, integer disk assignment and varying-background positivity remain global source work.
06	Primitive integral null class, finite chamber obstructions, curve-class alternatives and nonsplitting.	The positive period plane need not be rational. Torsion, integral multiplicities and all possible later curve classes are retained.
07	Restricted exact-pair transversality, normal deformation operator, nodal insertion and a proper torus pencil.	Exact variations with B fixed are not arbitrary changes of almost-complex structure. The quotient must be Hausdorff and smooth.

Module	Source mechanism	Boundary retained here
08	Periodic monopole, circle bundle, Hopf completion, fixed holomorphic family and exponential annular matching.	The full nodal model and its two period tests are imported; Proposition 5.1 is only an annular tail estimate.
09	Torus averaging, support-preserving exact homotopy, invariant preparation and prescribed endpoint tamers.	Fibrewise positivity and fixed global periods are both needed.
10	Semi-flat form, nodal gluing, three moment corrections, exact-form inverse, quadratic solve and uniform chamber bound.	The local radius is sharpened; collapse geometry, continuous inverse bounds and all-curve chamber estimate are not manufactured by finite tests.
11	Two Kähler volume constructions and an actual smooth path in the original three classes.	The auxiliary D is not substituted for the original B. Endpoint representatives and signs must match.

7.1 Three particularly delicate transitions

From currents to curves. Source Sections 3–4 retain angular information as well as mass. The density truncations are proved closed before integral-cycle regularity is invoked. In Section 5, compact-family curvature limits require an integer assignment on disks and positivity for the limiting almost-complex structure. Finitely many prescribed incidences, tangent directions, and approximation tolerances are chosen before the large Seiberg–Witten parameter. These steps cannot be replaced by positivity of a finite Gram matrix.

From local models to a global auxiliary solution. The annular difference has two relevant periods: fibre area and the swept invariant cycle. Matching just one does not make it exact. The source's bounded annular primitive is formed on a fixed annulus; the transported collapsing charts, not a fixed ambient atlas, provide the polynomial bounds. Once the three global moments are corrected, the exact-form inverse avoids a one-form spectral-gap estimate by Stokes. This valid distinction is preserved in Section 4. Our larger radius does not remove the embedding or positive-cone test.

From one auxiliary form to every future curve. The source writes $[D_\varepsilon] = \lambda_\varepsilon F + \varepsilon u_\varepsilon$ with $\lambda_\varepsilon \asymp \varepsilon^{-1}$, bounded u_ε, and controlled pairing $u_\varepsilon F$. The Lorentz decomposition of an integral curve class of square at least -2, together with integral Fd, gives one sufficiently small ε excluding all unwanted classes. It is chosen *before* the two Kähler steps. Checking a finite list of curves would not prove that uniform statement. The first step produces $C_1 \in [C]$, the second $B_1 \in [B]$; a smooth family of tamers is then chosen with the actual prescribed endpoints. None of our local error bounds independently establishes this chamber or endpoint construction.

8 Comparison, extensions, and current literature

Demonstrated local increment. The source already proves the spectral-gap-free exact inverse, polynomial fixed-order collapse control and exponential residual. We do not credit these as new. Theorem 2.1 turns its Stokes mechanism into an exact two-state comparison, retains the topological defect, and yields the sharp square-root density constant 4. Theorem 3.1 quantifies the price of

using a fixed physical norm. Theorem 4.1 replaces the displayed sufficient condition $\sigma < 1/2$ by $\sigma < 1$ without changing the equation or its three compatibility moments. The annular tail has explicit constants and a stated domain; its Bessel and geometric-series ingredients are classical.

Advantages and costs. The midpoint estimate is derivative-free, applies to the actual nonlinear volume difference, and has no one-form gap in its constant. Its cost is exact closedness, exact cohomology and orthogonality, together with a comparison metric depending on both forms. A fixed-background estimate pays e^{Ξ}. The radius test can certify larger residuals, but its contraction becomes slow near the discriminant endpoint and still needs genuine H^k inverse and multiplication bounds. The Fourier tail is useful for annular truncation and positivity, but loses uniformity if the annulus approaches the node on the scale of ρ, and does not preserve the completed geometry by itself.

Feasible extensions with missing hypotheses stated. For a compact parameter family, the same arguments give uniform estimates if P, Q, S, m, Ξ and the discriminant margin are uniformly controlled in the declared norms. Smooth parameter dependence requires the corresponding family regularity and derivatives; pointwise existence at each parameter is insufficient. For inexact orthogonality, the full three-slot formula (6.1) suggests an error budget with the actual inverse Gram matrix, provided the average span remains positive and the cohomological term is retained. For a manifold with boundary, Stokes contributes $\int_{\partial X} a\,da$; the closed-manifold formula must not be reused without boundary conditions controlling that term. An a posteriori collapse computation would additionally need certified radial and connection-primitive tails, all three global moments, and the continuous inverse/embedding constants. These are research tasks, not performed numerical experiments.

Primary literature status as of October 9, 2026. Donaldson's 2006 paper [2] supplies the classical negative-chord viewpoint and elliptic formulation; the relevant uniqueness proof and prescribed-volume example were read. Fine–He–Yao [3] prove the deformation result under an effective circle symmetry and identify the underlying manifold with T^4. The primary record gives version 2 of August 13, 2025 and publication in IMRN 2025, issue 16. Its abstract and version record, not the full proof, were checked here.

The current records for Petcu [4], version 1 of April 20, 2026, and Yao–Zhou [5], corrected version 3 of May 12, 2026, concern a coflow description and symmetry-dependent G_2 reductions, respectively. Their displayed abstracts do not establish the unrestricted deformation assertion of the assigned source. The source is a September 23, 2026 OpenAI Math Release preprint claiming that unrestricted result. Cached byte identity and public availability are provenance, not community validation or a substitute for verifying its global proof. No exhaustive historical novelty search or independent validation of the full external Taubes, current-regularity, or Yau proofs is claimed.

9 Evidence scope and conclusion

The public reading covers all 16 cached text files, 7,104 lines, including all 6,251 lines of the eleven mathematical modules, the main file, preamble, endpoint diagram, bibliography and README. Literal line intervals were recorded only after bounded successful displays; the PRE hashes were persisted before reading and the POST hashes are checked at finalization. A full text read is not a full proof audit. The source's external references and separately installed LaTeX infrastructure are not included in this coverage.

Fresh private proof reads comprise TR8 lines 63–180, the full 295-line stability file, and the jet file lines 1–160, 522–575, 747–865: 746 distinct lines. The exact unread complements are in the reading ledger. Included chapters and bibliography items outside those intervals were not silently counted as read. No whole-corpus claim is made, and no confidential corpus material was sent in public literature queries.

The owned executable check suite has **262 exact rational or symbolic cases and 1,461 assertions**. It tests Lorentz midpoint identities, physical boosts, mass-preserving density cells, the quadratic root, the cohomology negative control and the sharpness limit. Separately, **18 floating-point diagnostic cases and 36 assertions** compare Bessel partial tails and elementary majorants at 50 decimal digits. These are not interval enclosures; the infinite-tail inequality is established by its analytic proof. All counts exclude parent and other reports' checks. The tests do not certify global existence, current recognition, family invariants, the pencil, nodal completion or the all-curve chamber argument.

No correction to the source's stated normalization or equation is asserted by this report. The sharper radius is a refinement of a valid sufficient condition, not a repair of a false source theorem. The principal outcome is a quantitative comparison in the correct physical metric, with exact cohomology and positivity made indispensable. Together with the improved local correction test and bounded annular tail, it supports contribution level C3 for a notable source-specific local increment. This assessment is independent of applicability L3 and is not credit for the source's global headline or for classical uniqueness.

References

[1] OpenAI, *Deforming hypersymplectic four-manifolds to hyperkähler triples*, Math Release preprint, September 23, 2026. https://github.com/openai/math/tree/main/preprints/Deforming-hypersymplectic-four-manifolds-to-hyperkahler-triples-September-23-2026. All assigned cached text read; global claim treated as the source's claim.

[2] S. K. Donaldson, *Two-forms on four-manifolds and elliptic equations*, in *Inspired by S. S. Chern*, 2006, pp. 153–172. https://arxiv.org/html/math/0607083. Scoped proof reading: Section 2, Proposition 1, and the prescribed-volume discussion in Section 3.

[3] J. Fine, W. He and C. Yao, *Hypersymplectic structures invariant under an effective circle action*, IMRN 2025(16), rnaf252. https://arxiv.org/abs/2503.05272v2. Abstract and primary version/publication record checked.

[4] A. M. Petcu, *The hypersymplectic flow descended from the G_2-Laplacian coflow*, 2026. https://arxiv.org/abs/2604.18554v1. Abstract and version record checked; full proof not audited.

[5] C. Yao and Z. Zhou, *Closed G_2-structures with T^3-symmetry and hypersymplectic structures*, 2026. https://arxiv.org/abs/2601.13747v3. Abstract and corrected-version record checked; full proof not audited.

[6] NIST Digital Library of Mathematical Functions, equation 10.32.9, modified Bessel integral representation. https://dlmf.nist.gov/10.32.E9. The displayed formula and its domain were checked.

[7] M. Subbarao, private original Ram PDE manuscript, *TR8-lin-non-lin*, local file `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`, lines 63–180 freshly read. Confidential local reference.

[8] M. Subbarao, private original stability manuscript, `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, lines 1–295 freshly read. Coercivity, physical scaling, full state derivatives and residual-ball constraints; confidential local reference.

[9] M. Subbarao, private original derivative-jet manuscript, `RamNonlinearIE-PDE/TR1-lin-nonlin-rev2/sec-03-jets.tex`, lines 1–160, 522–575 and 747–865 freshly read. Exact Taylor-moment and derivative-remainder proofs; confidential local reference.